

SAVE, Benin

WMO: 653350

Lat: 8.033N

Lon: 2.467E

Elev: 656

StdP: 14.35

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	69.2	70.5	35.3	31.0	81.5	39.3	36.3	81.2	12.2	79.8	10.5	78.6	2.0	0	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	19.2	98.1	73.8	96.6	74.1	95.2	74.5	80.4	89.9	79.7	89.3	79.1	88.8	4.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
78.0	148.9	85.5	77.2	145.1	84.5	76.5	141.8	83.8	44.5	89.8	43.8	89.8	43.2	89.2	87.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.0	9.3	7.7	65.9	101.7	2.7	1.6	63.9	102.8	62.3	103.7	60.8	104.6	58.8	105.7		
(5)			51.3	82.8	1.9	2.1	49.9	84.3	48.8	85.5	47.8	86.7	46.4	88.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	82.9	86.0	86.1	84.3	82.4	80.2	78.7	77.6	78.9	80.4	82.9	82.8		
	DBStd	3.44	2.77	2.16	2.31	2.65	2.54	2.06	1.81	1.69	1.72	1.92	1.85	2.04		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDD50	11646	1020	1009	1121	1029	1003	907	888	856	866	942	987	1018		
	CDD65	6171	555	589	656	579	538	457	423	391	416	477	537	553		
	CDH74	65900	6828	7919	8575	7014	5882	4233	3204	2520	3128	4275	5869	6453		
	CDH80	28883	3461	4334	4581	3369	2447	1366	746	447	786	1500	2652	3194		
(14) Wind	WSAvg	3.0	2.5	3.1	3.6	3.9	3.4	3.3	3.3	3.6	2.8	2.2	2.0	2.3		
(15) Precipitation	PrecAvg	43.4	0.2	0.7	2.5	4.2	4.9	6.2	6.5	5.9	6.5	4.5	0.6	0.2		
	PrecMax	76.2	2.7	3.1	7.1	8.3	9.6	13.8	26.4	13.9	13.5	10.5	5.2	2.1		
	PrecMin	25.2	0.0	0.0	0.2	0.4	1.2	2.0	1.0	1.1	0.5	0.4	0.0	0.0		
	PrecStd	10.1	0.5	0.8	1.7	2.1	2.0	2.4	4.4	3.3	2.7	2.0	1.0	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.2	100.3	100.6	98.1	95.3	90.8	88.0	87.4	88.9	91.6	95.4	96.9		
		MCWB	70.3	72.4	74.6	77.7	78.1	77.7	77.0	76.7	77.5	77.9	74.3	69.2		
	2%	DB	95.8	97.9	98.2	95.7	92.6	89.2	86.5	85.4	87.2	89.7	93.4	94.6		
		MCWB	71.0	73.0	75.0	78.1	78.4	77.6	76.7	76.3	77.1	77.7	75.0	70.6		
	5%	DB	94.1	96.4	96.4	93.9	91.0	87.9	85.3	84.0	85.8	88.2	91.9	93.1		
		MCWB	71.5	73.5	75.3	78.1	78.4	77.5	76.4	75.9	76.9	77.6	75.3	71.5		
	10%	DB	92.5	95.0	94.6	92.2	89.6	86.5	84.0	82.5	84.4	86.8	90.5	91.7		
		MCWB	71.2	74.2	76.0	78.2	78.5	77.2	76.0	75.7	76.6	77.6	75.5	72.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.0	80.2	80.4	81.0	81.6	79.6	78.8	78.3	79.1	80.5	80.8	78.2		
		MCDB	90.4	92.9	92.0	91.3	89.8	87.6	85.6	84.3	86.0	86.8	87.9	90.1		
	2%	WB	77.5	78.9	79.2	80.0	80.1	78.7	77.7	77.2	78.1	79.2	79.2	77.2		
		MCDB	88.4	91.0	91.2	90.7	89.0	86.8	84.5	83.3	85.1	86.6	87.8	88.5		
	5%	WB	76.6	78.0	78.6	79.3	79.4	78.1	77.0	76.5	77.4	78.5	78.2	76.4		
		MCDB	87.0	89.5	90.5	90.0	88.4	85.9	83.8	82.4	84.3	86.2	87.4	86.8		
	10%	WB	75.8	77.3	78.0	78.8	78.7	77.4	76.3	75.9	76.8	77.9	77.6	75.7		
		MCDB	85.6	88.3	89.3	89.5	87.7	84.9	82.7	81.5	83.5	85.4	86.8	85.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	19.9	19.2	18.4	16.2	14.1	12.3	10.6	10.1	11.5	13.9	17.2	20.4		
		MCDBR	22.3	21.0	20.8	18.5	16.5	14.4	12.6	12.7	13.7	15.9	19.1	21.9		
		MCWBR	7.0	5.7	5.2	4.7	5.0	5.2	4.9	5.4	5.8	6.4	5.3	6.8		
		MCDBR	17.6	18.3	18.0	16.1	14.5	13.0	11.5	11.2	12.5	14.7	16.4	17.4		
	5% WB	MCWBR	4.7	4.7	5.0	5.3	5.7	5.4	5.1	5.4	5.9	6.7	5.7	4.7		
(40) Clear-Sky Solar Irradiance	taub		0.783	0.807	0.821	0.702	0.596	0.544	0.514	0.506	0.501	0.564	0.617	0.680		
	taud		1.534	1.515	1.486	1.708	1.903	2.048	2.159	2.175	2.191	1.980	1.815	1.697		
	Ebn at Noon		189	191	191	213	231	240	249	255	260	243	226	208		
	Edn at Noon		91	95	98	78	62	53	48	48	48	59	68	75		
(44) All-Sky Solar Radiation	RadAvg		1614	1646	1722	1724	1652	1492	1292	1167	1349	1555	1649	1594		
	RadStd		43	56	67	57	62	77	60	77	76	56	46	54		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.50	+0.56	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page